

Assignment D 42620/42622: Know how to use the atlas

Who is responsible for this assignment?

Our work has been split equally between all group members.

Introduction

You will prepare an analysis stemming from the following paragraph, part of statement id=4839 from *Kraka Economics*, on LinkedIn, the 2025-12-13:

“In October, the economic experts recommended dropping the Energy Island Bornholm project unless positive economic benefits for society could be demonstrated. At the same time, they called for an analysis of supply security, among other things. However, there is much evidence to suggest that this is not enough to turn the project's economic outlook positive.”

Remark that this statement is about (1) **energy supply security** and (2) **expertise**. You will gather data from the corpus to contextualize and contrast Kraka Economics' perspective. That data will aim at answering questions such as: Who is presented as an expert on questions related to energy supply security? What do these experts contend? Are they criticized, and by whom? Etc.

However, you will not actually produce the analysis. In this assignment, you will just **build queries and generate visualizations** that could potentially be used as source material for a policy brief. The assignment aims at checking that you know how to navigate the dataset and use the notebooks at your disposal.

Questions

Please paste your answers directly in the template below.

- 1) What are other formulations of the notion of “energy supply security”? We will use those to build a richer query.

Strategy: Ingest the corpus as Markdown files into a knowledge stack in Msty Studio or into NotebookLM and ask it to retrieve statements about energy supply security, read them, and identify other ways to talk about energy supply security

Alternative strategy: query the corpus for "supply security", read the statements to find other formulations, then use those as new queries, and iterate this way to find many formulations.

1. Geopolitical strategies and independence (212)
2. Technical reliability and grid stability (1896, 73)
3. Economic and market stability (85, 1817)

- 2) Build a query that captures the question of supply security. Provide a screenshot of the query and how many statements it finds in the corpus.

Strategy: compose a query with the formulations you have just identified.

find statements and include their numbers that use 'energy independence from Russia'
OR 'market stability' OR 'cyber and physical attacks'

Based on the sources, the following statements explicitly use the phrases you identified in your query:

Energy independence from Russia

Multiple actors, primarily Dan Jørgensen, emphasize that the energy islands are a strategic tool to achieve geopolitical autonomy:

- **Statement 212:** Dan Jørgensen notes that the expansion and connection to Germany is "absolutely essential for Europe's **energy independence from Russia**" 1 .
- **Statement 1059:** Dan Jørgensen repeats that the project is "essential for Europe's **energy independence from Russia**" 2 .
- **Statement 1100:** Dan Jørgensen highlights that the initiative is "absolutely essential for Europe's **energy independence from Russia**" 3 .
- **Statement 1720:** Dan Jørgensen states the connection is "absolutely essential for Europe's **energy independence from Russia**" 4 .
- **Statement 1720:** Dan Jørgensen states the connection is "absolutely essential for Europe's **energy independence from Russia**" 4 .
- **Statement 2739:** Dan Jørgensen asserts that the Bornholm project is "vital for Europe's **energy independence from Russia**" 5 .
- **Statement 3142:** Dan Jørgensen again defines the plan as "essential for Europe's **energy independence from Russia**" 6 .

Market stability

The term is used in technical contexts regarding grid mechanisms and the regulation of prices:

- **Statement 1180:** Peter K. Kruse-Andersen discusses the interaction between the energy island and the "**Market Stability** Reserve (MSR)" within the EU Emissions Trading System 7 8 .
- **Statement 5651:** A Reddit user argues that a pricing reform should prioritize affordable energy for consumers while ensuring "**market stability**" 9 .

Cyber and physical attacks

Security concerns regarding the vulnerability of offshore infrastructure mention these types of threats:

- **Statement 1896:** Anders Stouge asserts that energy islands "must be designed to withstand both **cyber and physical attacks**" ¹⁰ .
- **Statement 3870:** Melanie Markwick-Day highlights that modernization and digitalization introduce risks such as "vulnerability to cyberattacks and potential damage from **physical attacks** by malicious actors" ¹¹ .

3) Is there an issue that corresponds to energy supply security?

Strategy: read the issue descriptions.

Will Denmark be more independent from energy imports?

Will energy islands make the energy market more stable?

How will the new energy market impact people with lower income?

How will an added layer of technology impact the resilience of the energy grid?

4) Is there a cluster that corresponds to energy supply security?

Strategy: visualize by clusters using the query composed above.

Energy supply security can be grouped into three main clusters. The first cluster is geopolitical security, focusing on energy independence and reduced reliance on imports. The second cluster is grid resilience, covering the physical robustness and cybersecurity of energy infrastructure against disruptions and attacks. The third cluster is market stability, which influences public confidence and is particularly important for ensuring affordable energy access.

5) Build a query that captures statements about *both* energy supply security and expertise and test it. You should obtain at least 15 statements. If you do not have enough, improve your query by expanding on the vocabulary about either energy supply or expertise. Provide a screenshot of your query and how many statements it matches.

Strategy: combine your query about energy supply security with a query about expertise.

Paste screenshot here.

Query: ("energy security" OR "supply reliability" OR "resilience" OR "energy independence" OR "stable electricity supply") AND ("research" OR "analysis" OR "professor" OR "engineer" OR "expert" OR "competencies" OR "DTU")

Results: This query identifies at least **22 statements** in the provided corpus where experts (professors, researchers, and technical analysts) discuss supply security. Here are 15 specific examples:

1. **Kraka Economics (Senior Economist):** Their **analysis** notes that concerns about **power adequacy** (supply reliability) are decreasing **4** .
2. **Professor Jacob Østergaard (DTU):** Warns that **research** is needed to ensure energy islands do not compromise the standard of **supply reliability** **3** .
3. **Lars Aagaard (Minister/Expert context):** Outlines technical ways to ensure a **stable electricity supply** through grid integration **5** .
4. **Professor Brian Vad Mathiesen (Aalborg University):** Identifies **energy security** and cost efficiency as the key factors for moving the project forward **8** .
5. **Anders Stouge (DTU Researcher):** Asserts the system must be designed with **resilience** to withstand cyber and physical attacks **9** .

6. **Jørgen S. Christensen (Green Power Denmark CTO):** Argues that energy hubs must be **resilient** to future disruptions 10 .
7. **DTU Sector Development Project:** Recommends **research** into securing **resilience** against digital and physical threats 11 .
8. **Danish Council on Climate Change:** Provides an **analysis** showing Denmark can maintain high **electricity supply security** through flexible consumption 12 .
9. **Arash July (DTU):** **Research** into grid-forming control techniques to ensure the **stable** operation of energy islands 13 .
10. **Peter K. Kruse-Andersen (University of Copenhagen):** **Research** on the **Market Stability Reserve** and EU Emissions Trading 14 .
11. **Fraunhofer IEE (Researchers):** Their **study** shows that energy islands improve **system resilience** and save grid costs 15 .
12. **N. Pardo García (AIT Researcher):** **Analysis** on increasing panel capacity and battery storage to mitigate **blackout** risks 16 .
13. **Rocky Mountain Institute (Researchers):** Their **study** argues that island communities enjoy improved **reliability and stability** of power 17 .
14. **Artur Usanov & Alexander Kharin (Researchers):** Scientific **analysis** on cost-effective ways to ensure grid **reliability** 18 .
15. **Claire Bergaentzlé (DTU Energy Economist):** **Analysis** showing how failed tenders jeopardize renewable targets and **grid goals** 19 .
- What to use for your exercise:** You can provide the **Boolean string** above and state that it successfully retrieves more than the required **15 statements** (specifically identifying at least 22) where technical expertise is applied to the question of supply security.

- 6) Find at least 3 statements where you have verified that a person or an organization has been presented as an expert in an argument related to energy supply security. Take note of (1) the id of the statement, (2) who is the author of the statement, (3) who is the expert, and (4) interesting things you remark on the way (optional).

Strategy: export the corpus as text using the query above and read the statements. Some of them, possibly many, will be false positives: you will have to read more than 3 to find 3 statements.

One statement per row. Use this format:

id; statement author; mentioned expert; optional comment

8, Steffen Olsen (Kärnfull Energi), Frederik Læssøe Nielsen (Kraka Economics)

1273, Facebook user (General public opinion), P-FB (Paul-Frederik Bach), an energy consultant.

380, BBC, Jacob Østergaard

290, Ingeniøren, Jan Bentzen

- 7) The Danish Energy Agency is presented as an expert but also criticized. Visualize the evolution of the proportion of statements mentioning the agency so that we can identify when the peak is reached. Set your visualization to make the pattern as visible as possible, and highlight it with an annotation.

Paste screenshot of the annotated visualization.

- 8) Visualize the evolution of the proportion of statements authored by the agency so that we can identify when the peak is reached. Set your visualization to make the pattern as visible as possible and highlight it with an annotation.

Paste screenshot of the annotated visualization.

- 9) There are 15 statements by Kraka Economics mentioning “Energy Agency”. What type(s) of sources do they come from? Provide your answer as a screenshot of a visualization featuring the exact number of statements per source type (no annotation needed).

Paste screenshot of the visualization.

Evaluation criteria

	Good (3 pts)	Approved (2 pts)	Not approved (1 pt)
Question 1	3 query examples related to energy supply security provided.	3 query examples provided, but they are not related to energy supply security.	3 examples of queries not provided.
Questions 2	Screenshot of demanded content provided.	Screenshot of demanded content provided but with mistakes.	Screenshot with demanded content not provided.

Questions 3	Correct issue(s) / cluster(s) identified.	Correct issue(s) / cluster(s) identified but with mistakes.	Correct issue(s) / cluster(s) not identified.
Questions 4	Correct issue(s) / cluster(s) identified.	Correct issue(s) / cluster(s) identified but with mistakes.	Correct issue(s) / cluster(s) not identified.
Questions 5	Screenshot of demanded content provided.	Screenshot of demanded content provided but with mistakes.	Screenshot with demanded content not provided.
Question 6	3 statements provided.	3 statements provided but with mistakes.	3 statements not provided.
Questions 7	Annotated visualization with demanded content provided.	Right content provided, but annotation missing, or wrong choice of visualization.	Annotated visualization with demanded content not provided.
Questions 8	Annotated visualization with demanded content provided.	Right content provided, but annotation missing, or wrong choice of visualization.	Annotated visualization with demanded content not provided.
Questions 9	Screenshot of demanded content provided.	Screenshot of demanded content provided but with mistakes.	Screenshot with demanded content not provided.

Points obtained in Assignment D translate in the following way to overall performance:

	Good	Approved	Not approved
Overall performance	Minimum 23 pts	Minimum 18 pts	Minimum 0 pts